25900784 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-BCA (HONOURS) REGULAR /IMPROVEMENT/ REAPPEARANCE EXAMINATION NOVEMBER 2025

FIRST SEMESTER

Core Course (CC) - MG1CCRBCA101 - DISCRETE MATHEMATICS

(2024 ADMISION ONWARDS)

Duration: 2 Hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer Type Questions

Answer any **5** Questions

Each Question carries **2** marks

1. Identify which rule of inference is the basis of the following argument: It is below freezing now. Therefore, it is either below freezing or raining now. [A] / [CO5]

2. Define a bit string. Give a bit string of length nine. [K] / [CO3]

3. Define $A \cap B$. If $A \subseteq B$ then find $|A \cap B|$? [K] / [CO2]

4. Define the cartesian product of two sets. [K] / [CO1]

5. Define a directed graph with an example. [K] / [CO6]

6. Let R_1 and R_2 be the relations from $A = \{1, 2, 3\}$ to $B = \{1, 2, 3, 4\}$ where $R_1 = \{(1, 1), (2, 2), (3, 3)\}$ and $R_2 = \{(1, 1), (1, 2), (1, 3), (1, 4)\}$. Find $R_1 \cap R_2$ and $R_1 - R_2$. [A] / [CO6]

7. Show that the matrix $A = \begin{bmatrix} 5 & 3 \\ -1 & 2 \end{bmatrix}$ is non-singular. [U] / [CO7]

8. For the matrices $A = \begin{bmatrix} 2 & 3 \\ 1 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} 4 & 2 \\ 1 & 3 \end{bmatrix}$, find $6A + 3B$. [U] / [CO7]

[5x2 = 10]

Part B

Short Essay Type Questions

Answer any **5** Questions

Each Question carries **6** marks

9. Construct the truth table for the compound proposition $(p \to q) \to (q \to p)$. [U] / [CO3]

10. Explain the following. (a) Proposition (b) Propositional variables (c) Propositional logic [K] / [CO3]

11. List the members of these sets. [U] / [CO1]
    (a) $\{x : x \text{ is a positive integer less than} 12\}$
    (b) $\{x : x \text{ is an integer such that } x^2 = 2\}$
    (c) $\{x : x \text{ is an odd positive integer less than } 10\}$

12. Define a graph of a function. Display the graph of the function $f(x) = x^2$ from the set of integers to the set of integers. [K] / [CO6]

13. Check whether the following relation $\{1, 2, 3, 4\}$ is symmetric and/or antisymmetric where, $R = \{(1, 1), (1, 2), (2, 1), (2, 2), (3, 4), (4, 1), (4, 4)\}$. [A] / [CO6]

14. Let $R = \{(a, b)/\text{a and b speak a common language}\}$ is a relation on the set of all people. Is it an equivalence relation? Justify your answer. [An] / [CO6]

15. Solve the following system of equations using Cramer's rule: [A] / [CO8]
    $3a + b + 2c = 3$
    $2a - 3b - c = -3$
    $a + 2b + c = 4$

16. Find the inverse of the matrix $A = \begin{bmatrix} 1 & 3 & 3 \\ 1 & 4 & 3 \\ 1 & 3 & 4 \end{bmatrix}$. [A] / [CO8]

[5x6 =30]

Part C

Essay Type Questions

Answer any **3** Questions

Each Question carries **10** marks

17. (a) Define a tautology. (b) Prove that $(\neg p \wedge (p \to q)) \to \neg q$ is a tautology. [A] / [CO4]

18. Describe the logical operations of conjunction, disjunction, and exclusive or between two propositions using examples, and present their truth tables. [U] / [CO3]

19. Let $A = \{0, 2, 4, 6, 8, 10\}, B = \{0, 1, 2, 3, 4, 5, 6\}$ and $C = \{4, 5, 6, 7, 8, 9, 10\}$. Find $A \cup B \cup C$ and $A \cap B \cap C$.
    Also, draw the Venn diagrams for these two combinations. [A] / [CO2]

20. Let R be the relation on the set of integers. R is defined as follows: [A] / [CO6]

$a\,R\,b \iff a \equiv b \pmod 3.$

(a) Show that R is an equivalence relation.

(b) Find the equivalence classes under this equivalence relation and check whether it is a partition of the set of integers.

21. If $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 4 \\ -1 & 1 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} 0 & 2 & -1 \\ 1 & 3 & 4 \\ 0 & -2 & -3 \end{bmatrix}$, verify that $(AB)^T = B^T A^T$. [U] / [CO7]

22. (a) Solve the following system of equations using the matrix method: [A] / [CO8]

$a + b + c = 3$

$a + 2b + 3c = 4$

$a + 4b + 9c = 6$

(b) If $A = \begin{bmatrix} 3 & 1 \\ -1 & 2 \end{bmatrix}$, show that $A^2 - 5A + 7I = 0$, where I is the identity matrix of order 2.

[3x10 =30]